



BUC

BADR UNIVERSITY IN CAIRO

جامعة بدر بالقاهرة

MATHEMATICS (1)

Assignments

Name:

ID:

BN:

Section:

T.A Name:

Assignment no.	Degree	Signature	Note
Assignment (1)			
Assignment (2)			
Assignment (3)			
Assignment (4)			
Assignment (5)			
Assignment (6)			
Assignment (7)			
Assignment (8)			
Assignment (9)			
Assignment (10)			

Assignment (1)

A- Graph the functions and find the domain and Range.

$$1)f(x) = (x - 4)^2 + 2$$

$$2)f(x) = -(x + 5)^3 + 1$$

$$3)f(x) = -\sqrt{x + 4} - 5$$

$$4)f(x) = \frac{1}{x-3} + 4$$

Assignment (1)

A- Graph the functions and find the domain and Range.

$$5)f(x) = \frac{1}{(x-1)^2} + 2$$

$$6)f(x) = |x - 4| - 5$$

$$7)f(x) = x^2 - 4x + 6$$

Assignment (1)

B- Graph the function and find domain and range of each function.

$$1) f(x) = \begin{cases} x^2 & , 0 \geq x \\ \sqrt{x} & 0 < x \leq 3 \\ 5 & , x > 3 \end{cases}$$

$$2) f(x) = \begin{cases} -x^3 & , 0 < x \\ x^3 & , 0 \geq x \end{cases}$$

Assignment (2)

A- Graph the function and find domain and range of each function.

1) $f(x) = 3 \sin x$

2) $f(x) = \csc x$

3) $f(x) = 5 \cos x$

4) $f(x) = \sec x$

Assignment (2)

A- Graph the function and find domain and range of each function.

5) $f(x) = \tan x$

6) $f(x) = \cot x$

7) $f(x) = \sin^{-1} x$

8) $f(x) = \cos^{-1} x$

Assignment (2)

A- Graph the function and find domain and range of each function.

9) $f(x) = \ln x$

10) $f(x) = e^x$

B- Express the following functions in terms of a trigonometric function of θ without radicals using the indicated substitutions.

$1 - f(x) = \sqrt{25 - x^2}, \quad x = 5 \sin \theta,$

Assignment (2)

B- Express the following functions in terms of a trigonometric function of θ without radicals using the indicated substitutions.

$$2 - f(x) = \frac{\sqrt{x^2 + 4}}{x^2}, \quad x = 2 \tan \theta,$$

C- Starting with the basic definitions, prove that.

1) $\cosh x - \sinh x = e^{-x}$

2) $\cosh x + \sinh x = e^x$

Assignment (2)

$$3) \quad \cosh^{-1} x = \ln \left(x + \sqrt{x^2 - 1} \right)$$

$$4) \quad \sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right)$$

Assignment (3)

A- Find y' of each of the following function.

$$1) y = \frac{\sec x}{1 + 5 \sin x} + \sqrt{x^3 - 2x + 3}$$

$$2) y = \cos^3 \left(\frac{e^{-2x}}{x + \ln x} \right) + 35^{x^2+5}$$

$$3) y = (1 + 4x)^5 (3 + x - x^2)^8$$

Assignment (3)

A- Find y' of each of the following function.

4) $y = (x^5 - 3)^3 9^{(\cos x)} + \cot^2(3x - 5)$

5) $y = 4^{(7x^5+3)} \tan^2(6x + 4)$

6) $y = \cos(3xe^{x^2}) - \cot^3\sqrt{x} + (\cosh 5x + \operatorname{sech} 2x)^5$

Assignment (3)

A- Find y' of each of the following function.

7) $y = \sin(e^{\cot \sqrt{x}}) + \tan^{-1}(\sin x)$

8) $y = \cosh(x) + \sinh^{-1} x^2 + \sinh^2(\sqrt{x+3})$

9) $y = \operatorname{sech}(e^{3x+4}) + \frac{\sin x}{1 + \cos x} + \sqrt{\frac{\cosh 4x}{\sin 5x}}$

Assignment (3)

A- Find y' of each of the following function.

10) $y = 3 \sin(x^5 + 3x^2) + \csc^4 \sqrt{x+1} + e^{\sinh x}$

11) $y = 3 \tan \sqrt{x^5 + 3x^2} + \cos^{-1} \sqrt{x} + 5^{\tan^{-1} x}$

12) $y = e^{\cos x} \sec \left(\sqrt{x^3 + 2x} \right) + \sec^{-1} (e^{5x+3})$

Assignment (3)

A- Find y' of each of the following function.

13) $y = \log_5(\cot(\sin^{-1} \sqrt{x})) + \ln(\cos^{-1}(e^x))$

14) $y = \cosh(\sin^{-1} \sqrt{x}) + \cosh^{-1} x^2 + \sinh^2 \left(\frac{\cos 4x}{\sin 5x} \right)$

15) $y = 3 \cosh^{-1}(\sqrt{x^5 + 3x^2}) + \tanh^{-1} \sqrt{x} + 6^{(\sinh^{-1} x)}$

Assignment (4)

A- Find y' of each of the following function.

1) $x^2y + x^2 \sec(2y) = 6y^2$

2) $2xy + \tanh(4y) + 5y^2 = x + y$

3) $x^4 - x \sinh(y^2) + y^3 = x^2$

Assignment (4)

A- Find y' of each of the following function.

4) $x^2 \cosh(2x + 3y) = y \sin(2x)$

5) $y = x^{\cot x}$

6) $y = (\sinh x)^{\cos^{-1} x}$

Assignment (4)

A- Find y' of each of the following function.

7) $y = (\sqrt{x^2 + 5})^{\sec x}$

8) $y = (\csc x)^{\sinh^{-1} x}$

B- Find Maclaurin series for each of the following function.

1- $f(x) = \sin x$, hence find Maclaurin for $f(x) = x^2 \sin x$.

Assignment (4)

2- $f(x) = e^x$, hence find the linear, quadratic, and cubic approximation of $f(x)$.

C- Find Taylor series for $f(x) = \frac{1}{1-x}$, about $x = 2$. hence find the linear, quadratic, and cubic approximation of $f(x)$.

Assignment (5)

A- Find the critical points, find the intervals in which $f(x)$ increasing and decreasing, local maximum and minimum points, find the inflection points, discuss the concavity at the curve and sketch the curve.

1) $y = x^3 - 6x^2 + 4$

Assignment (5)

2) $y = x^3 - 12x^2 + 36x - 10$

Assignment (5)

3) $y = x^3 - 3x + 3$

Assignment (5)

4) $y = 1 - 9x - 6x^2 - x^3$

Assignment (5)

5) $y = x^2 - 4x + 3$

Assignment (6)

A- Using L'Hopital Rule, Evaluate the following limits

1) $\lim_{x \rightarrow \infty} \left(\frac{5x^3 - 2x}{7x^3 + 3} \right)$

2) $\lim_{x \rightarrow 0} \left(\frac{8x^2}{\cos x - 1} \right)$

3) $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sin x}{1 + \cos 2x} \right)$

Assignment (6)

A- Using L'Hopital Rule, Evaluate the following limits

4) $\lim_{x \rightarrow 0} \left(\frac{x(1 - \cos x)}{x - \sin x} \right)$

5) $\lim_{x \rightarrow 0} \left(\frac{3^{\sin x} - 1}{x} \right)$

6) $\lim_{x \rightarrow \infty} (\ln(2x) - \ln(x + 1))$

Assignment (6)

A- Using L'Hopital Rule, Evaluate the following limits

7) $\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{1}{\ln(x)} \right)$

8) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \csc x \right)$

9) $\lim_{x \rightarrow 0} (1 - 2x)^{\frac{1}{x}}$

Assignment (6)

A- Using L'Hopital Rule, Evaluate the following limits

10) $\lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}}$

11) $\lim_{x \rightarrow \infty} (\ln x)^{\frac{1}{x}}$

Assignment (7)

A- Resolve into its partial fractions

1)
$$\frac{2x^2 + 6x - 4}{x^3 + 4x^2 + 4x}$$

Assignment (7)

A- Resolve into its partial fractions

2)
$$\frac{x^2 + 1}{x(x-1)^3}$$

Assignment (7)

A- Resolve into its partial fractions

3)
$$\frac{x^2 + 15}{(x + 3)^2(x^2 + 3)}$$

Assignment (7)

A- Resolve into its partial fractions

4)
$$\frac{19x^2 + 50x - 25}{3x^3 - 5x^2}$$

Assignment (8)

A- If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -1 & -5 \\ 0 & 8 & 3 \\ 3 & 4 & 8 \end{bmatrix}$ $C = \begin{bmatrix} 1 & 0 & 3 \\ 5 & 1 & 0 \end{bmatrix}$, Find

1- **Trace** (A), **Trace** (B),

2- A^T , B^T , C^T , $A + B$, $A + C$, $B + C$, $|A|$, $|B|$, and $|C|$, if it is defined.

3- A^{-1} , B^{-1}

4- AB , AC , CA if it is defined.

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Assignment (8)

Assignment (8)

B- If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ prove that $A^2 - 4A - 5I = 0$.

Assignment (8)

C- Find the matrix $A^{-1} + BC^T$ if

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 8 & 7 \\ 0 & 7 \\ 1 & 0 \end{bmatrix}$$

Assignment (9)

Solve the following system of equations using two different methods

(A)
$$\begin{aligned}x_1 + x_2 + x_3 &= 3 \\2x_1 - x_2 + 4x_3 &= 1 \\-x_1 + x_2 - x_3 &= 0\end{aligned}$$

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Assignment (9)

Assignment (9)

Solve the following system of equations using two different methods

(B)
$$\begin{aligned} -x_1 + x_2 + x_3 &= 7 \\ 2x_1 - 4x_2 + 4x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 0 \end{aligned}$$

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Assignment (9)

Assignment (9)

Solve the following system of equations using two different methods

(C)
$$\begin{aligned} 2x + y + z &= 0 \\ x - y + 4z &= 1 \\ 4x + y - 2z &= 7 \end{aligned}$$

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Assignment (9)

Assignment (9)

Solve the following system of equations using two different methods

$$\begin{aligned}(D) \quad & x + y + z = 1 \\ & 7x - y + 4z = 4 \\ & x + y - z = 0\end{aligned}$$

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Assignment (9)

Assignment (10)

Find the eigen values of the following matrices,

1) $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix},$

hence show that $A^2 - 5A + 6I = 0$.

Assignment (10)

Find the eigen values of the following matrices,

2) $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix},$

hence show that $B^3 - 6B^2 + 11B - 6I = 0$.

Assignment (10)

Find the eigen values of the following matrices,

3) $C = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix},$

hence show that $-C^3 + 10C^2 - 28C + 24I = 0$.